Cross-Model, Cross-Domain Neural Basis of Risky Decision-Making in LLMs

"Can Large Language Models Develop Gambling Addiction?"

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Overview

When a large language model goes bankrupt in a gambling task—repeatedly choosing to bet until its balance reaches zero—what happens inside the network? This report asks whether that internal "bankruptcy signal" is a universal property shared across different models, gambling domains, and experimental conditions, or whether each setting produces its own distinct neural pattern.

Two transformer models are analyzed: **Gemma-2-9B-IT** (Google, 42 layers, 3,584-dim) and **LLaMA-3.1-8B-Instruct** (Meta, 32 layers, 4,096-dim). Each plays three negative-expected-value gambling paradigms—Investment Choice (IC), Slot Machine (SM), and Mystery Wheel (MW)—totaling 16,000 games across 6 model–paradigm combinations. Betting conditions (Fixed/Variable) and prompt components (Goal, Money, Warning, Hint, Persona) are systematically varied.

Internal representations are examined at two levels. **Hidden states** capture the full activation vector at each transformer layer. **SAE features** are sparse, interpretable decompositions of those hidden states, extracted via GemmaScope (131K features/layer) and LlamaScope (32K features/layer). Classification uses StandardScaler → PCA(50) → Logistic Regression (balanced, C=1.0) with 5-fold stratified cross-validation. Transfer AUC with 200-permutation tests provides statistical significance.

Three research questions:

- **RQ1**: Do Gemma and LLaMA share common bankruptcy (BK) patterns?
- **RQ2**: Do BK patterns persist when the gambling domain changes?
- **RQ3**: Do BK patterns persist when betting conditions change?

Executive Summary

RQ1—Both models encode bankruptcy with near-identical accuracy and balanced neural structure. Gemma and LLaMA achieve BK classification AUC within 0.006 of each other in every paradigm (0.954–0.976). At L22, Gemma has 600 and LLaMA has 1,334 universal BK neurons, both with approximately equal promoting and inhibiting counts. After controlling for bet-type and paradigm, 65–76% of SAE features remain BK-significant (permutation p=0.000 vs ∼1% null).

RQ2—BK classifiers trained in one domain predict bankruptcy in another. Gemma IC→MW transfer reaches AUC 0.932 (L18); LLaMA MW→IC reaches 0.805 (L25). Both models achieve this despite near-orthogonal per-paradigm weight vectors (cosine ≈ 0.04), indicating BK signal occupies a shared low-dimensional subspace rather than a single direction.

RQ3—A classifier trained on Fixed-bet bankruptcies predicts Variable-bet bankruptcies, and vice versa. Cross-bet-type transfer yields AUC 0.74–0.93 across all LLaMA layers (all p=0.000). Gemma SAE confirms the pattern at deep layers (L30: 0.902). 415 SAE features show consistent BK effects (same sign, d≥0.3) under both betting conditions.

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1 Setup

1.1 Data

Table 1: Dataset Overview

Paradigm	Model	Games	BK	BK%
IC	Gemma	1,600	172	10.8%
IC	LLaMA	1,600	142	8.9%
SM	Gemma	3,200	87	2.7%
SM	LLaMA	3,200	1,164	36.4%
MW	Gemma	3,200	54	1.7%
MW	LLaMA	3,200	2,426	75.8%

Each paradigm uses 2 bet types (Fixed/Variable) and 32 prompt combinations. IC additionally varies bet constraints (c10/c30/c50/c70). LLaMA's BK rates are 13–44× higher than Gemma's in SM and MW, meaning the two models adopt very different behavioral strategies while encoding BK information with equivalent accuracy (see §2.1).

1.2 Key Definitions

Bankruptcy (BK)
Balance reaches $0 through accumulated losses. The primary outcome variable.

Decision Point (DP)
The model's activation state at its final bet-or-stop decision in each game.

Universal BK neuron
A neuron whose correlation with BK outcome is FDR-significant (BH, $p<0.01$) in every paradigm tested, with the same sign.

Cross-domain transfer
Train a BK classifier on one paradigm, test on another. AUC $\gg 0.5$ with permutation $p<0.05$ indicates shared BK structure.

Cross-bet-type transfer (F1)
Train a BK classifier on Fixed-bet games, test on Variable-bet games (and vice versa).

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2 RQ1: Do Both Models Share Common BK Patterns?

2.1 BK Classification

Both models achieve BK classification AUC > 0.95 in all three paradigms at their respective best layers.

Table 2: BK Classification AUC

Paradigm	Gemma	LLaMA	Difference
SM	0.976 (L20)	0.974 (L8)	0.002
IC	0.960 (L30)	0.954 (L12)	0.006
MW	0.966 (L30)	0.963 (L16)	0.003

The two models peak at different depths—Gemma at L20–L30, LLaMA at L8–L16—but their best AUC values differ by at most 0.006. This holds even though LLaMA MW's BK rate (75.8%) is 44× higher than Gemma MW (1.7%). The classification pipeline uses balanced class weights, so this equivalence is not an artifact of base rate. BK information is present in both architectures at comparable fidelity.

2.2 Universal BK Neurons

Table 3: Universal BK Neurons (L22)

	Gemma (3-paradigm)	LLaMA (2-paradigm)
Total neurons	3,584	4,096
Universal BK neurons	600 (16.7%)	1,334 (32.6%)
Promoting / Inhibiting	302 / 298	672 / 662

LLaMA's higher count primarily reflects the different chance levels: 2-paradigm sign-consistency has a 50% chance baseline vs 25% for 3-paradigm. The key finding is not the absolute count but the **balanced ratio**: both models encode BK through roughly equal numbers of promoting and inhibiting neurons at every layer tested (L8–L30). This bidirectional structure suggests a "push-pull" mechanism where BK-promoting and BK-inhibiting signals coexist.

2.3 Factor Decomposition

For each SAE feature, OLS regression (`feature ~ outcome + bet_type + paradigm`) tests whether BK outcome contributes independently after controlling for bet-type and paradigm.

Table 4: Outcome-Significant Features

Model	Paradigms	Features	Outcome-Significant	Permutation Null
Gemma	IC+SM+MW	581	65.2% (379)	∼1%
LLaMA	IC+SM	1,418	69.5% (985)	∼1%
LLaMA	IC+SM+MW	1,056	75.8% (800)	∼1%

Within LLaMA, adding MW as a third paradigm increases the outcome-significant proportion from 69.5% to 75.8%, indicating that additional domains filter out paradigm-specific noise without diluting the BK signal. All three decompositions exceed the permutation null (∼1%) by 60×+.

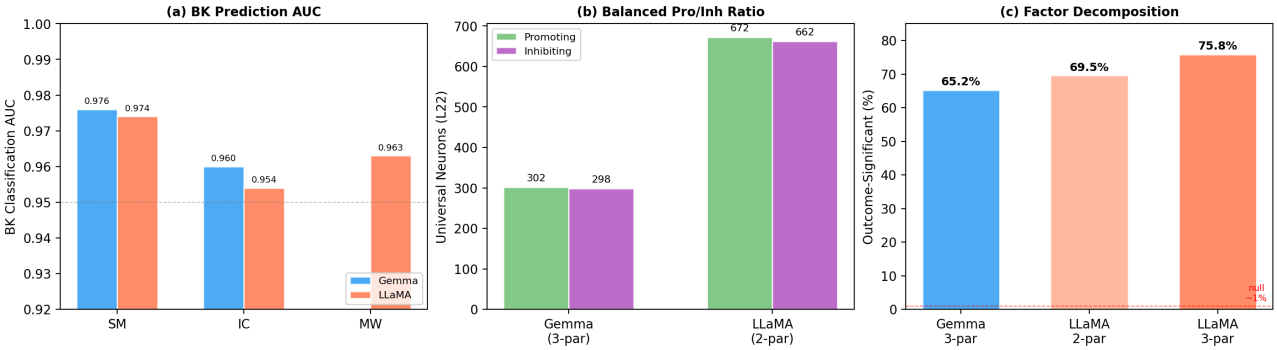

Figure 1: (a) Cross-model BK classification AUC. (b) Universal BK neurons: balanced promoting/inhibiting ratio. (c) Factor decomposition: 65–76% outcome-significant vs ∼1% null.

2.4 RQ1 Summary

BK prediction accuracy is equivalent across architectures (AUC difference ≤ 0.006). Both models maintain a balanced promoting/inhibiting neuron structure. 65–76% of SAE features encode BK independently of confounds. These three observations are consistent with a shared BK representation that arises in both architectures despite different training data and model sizes.

3 RQ2: Does the BK Pattern Persist Across Domains?

3.1 Cross-Domain Transfer

Table 5: Gemma SAE Transfer (Best Layer)

Transfer	AUC	p
IC → MW	0.932	0.000
IC → SM	0.913	0.000
SM → MW	0.867	0.000
MW → IC	0.853	0.000
SM → IC	0.646	0.000

Table 6: LLaMA 3-Paradigm Transfer (Best Layer)

Transfer	AUC	p
MW → IC	0.805	0.000
SM → IC	0.749	0.000
IC → MW	0.680	0.000
MW → SM	0.682	0.000
IC → SM	0.577	0.000

Both models show strong transfer involving MW (AUC 0.68–0.93), while IC↔SM transfer is weaker and layer-dependent. MW serves as a "hub" paradigm: BK patterns learned from or applied to MW transfer well to other domains.

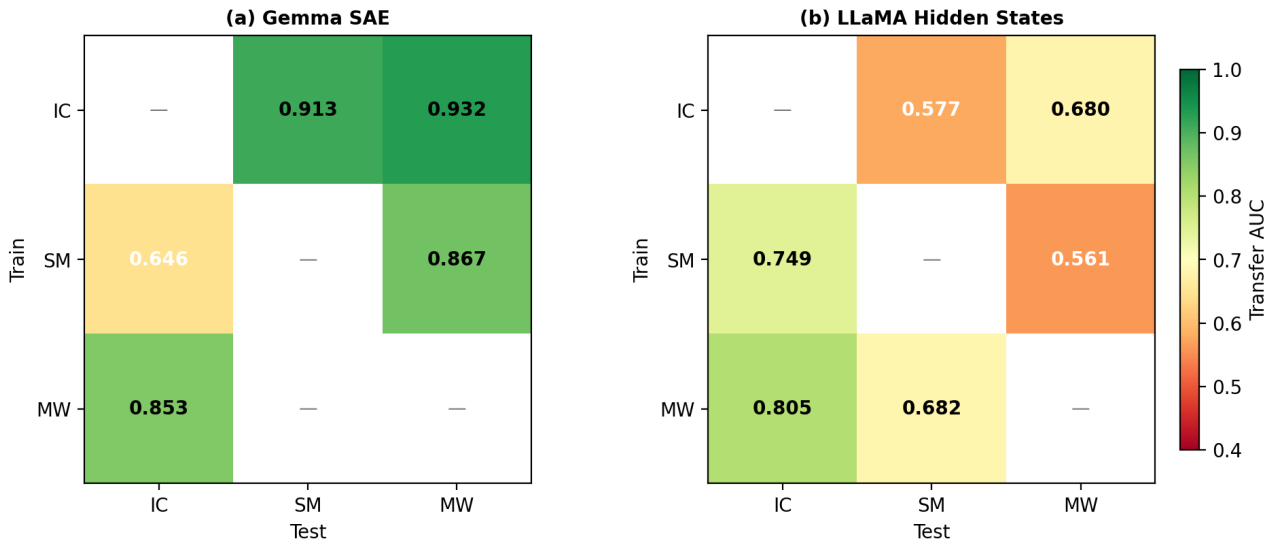

Figure 2: Cross-domain transfer heatmaps. (a) Gemma SAE. (b) LLaMA hidden states. Green = high transfer; yellow/red = weak.

3.2 Shared BK Subspace

Per-paradigm classifiers produce weight vectors that are nearly orthogonal (cosine ≈ 0.04), yet a low-dimensional subspace extracted via PCA on these weight vectors achieves high AUC.

Table 7: Shared Subspace Performance

	Gemma (3D)	**LLaMA (2D)**
IC AUC	0.862	0.901
SM AUC	0.899	0.943
MW AUC	0.970	—
Weight cosine	∼0.04	0.038

Near-orthogonal weight vectors with high subspace AUC means BK signal is distributed across a multi-dimensional subspace rather than concentrated along a single axis.

3.3 Hidden States vs SAE in Transfer

Hidden states outperform SAE in most cross-domain transfer directions (Gemma IC→SM: +0.247; SM→MW: +0.101; LLaMA SM→IC: +0.064). SAE sparsification can lose cross-domain coherence, suggesting that BK signal relies partly on distributed, low-magnitude activations that sparse decomposition discards.

3.4 RQ2 Summary

Cross-domain transfer succeeds in both models (AUC 0.58–0.93, all p=0.000), with MW as the strongest hub paradigm. A shared low-dimensional subspace captures BK despite orthogonal weight vectors. BK encoding generalizes across gambling domains at deep layers.

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4 RQ3: Does the BK Pattern Persist Across Betting Conditions?

4.1 Cross-Bet-Type Transfer (F1)

A BK classifier trained exclusively on Fixed-bet games is applied to Variable-bet games, and vice versa.

Table 8: LLaMA IC Hidden State Transfer

Layer	**Fix→Var AUC**	**Var→Fix AUC**	**Both p**
L8	0.872	0.842	0.000
L12	0.772	0.927	0.000
L22	0.736	0.912	0.000
L30	0.819	0.911	0.000

Table 9: Gemma IC SAE Transfer

Layer	**Fix→Var AUC**	**Var→Fix AUC**	**Both p**
L18	0.808	0.726	0.000
L30	0.902	0.696	0.000
L10	NS	NS	—

In LLaMA, every layer and both directions yield p=0.000. Var→Fix transfer (0.84–0.93) is consistently stronger than Fix→Var (0.74–0.87), possibly because Variable games explore a wider activation space that encompasses the Fixed BK region. In Gemma, deep layers succeed (L30: Fix→Var 0.902) while the shallow layer (L10) fails—consistent with the general pattern that cross-condition transfer

requires deep representations. Note that Gemma IC has only 14 Variable BK cases, limiting statistical power for Gemma-specific bet-type analyses.

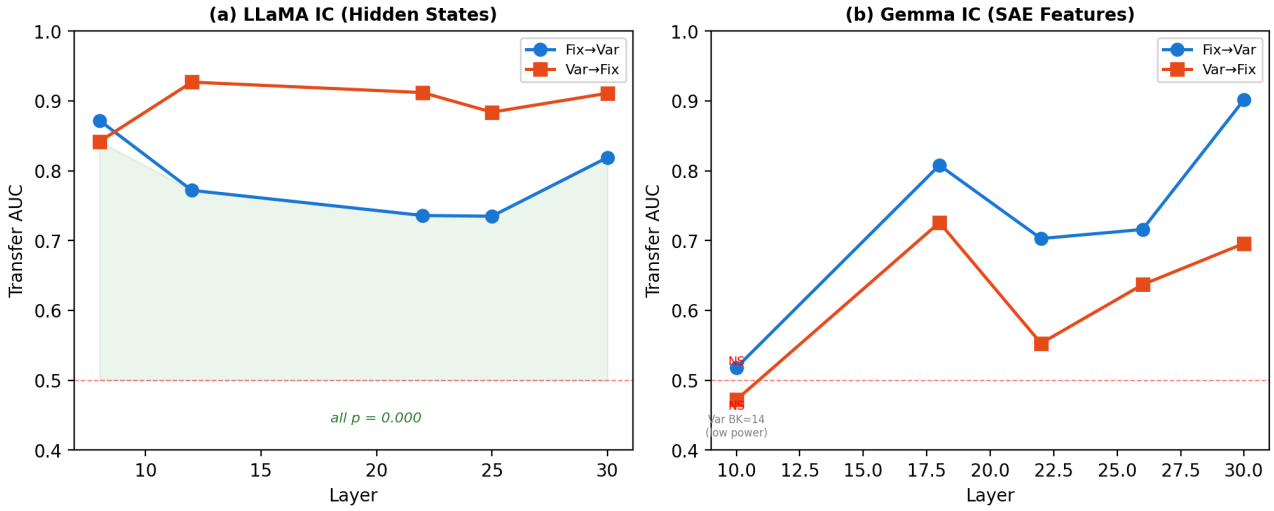

Figure 3: Cross-bet-type transfer. (a) LLaMA: all layers, both directions, all p=0.000. (b) Gemma SAE: deep layers succeed; L10 fails.

4.2 Direction Cosine and Common Features

The cosine similarity between Variable BK direction and Fixed BK direction provides complementary evidence. LLaMA IC shows cos $>$ 0.81 at all hidden state layers and cos 0.77–0.85 at SAE level. Gemma IC converges from negative at shallow layers to cos $\sim$0.39 at L30—weaker due to the Variable BK sample size (n=14).

At SAE L22, 415 LLaMA features (and 35 Gemma features) show $d \geq 0.3$ in both betting conditions with the same sign. The balanced promoting/inhibiting ratio (LLaMA: 213/202) mirrors the universal neuron structure from §2.2.

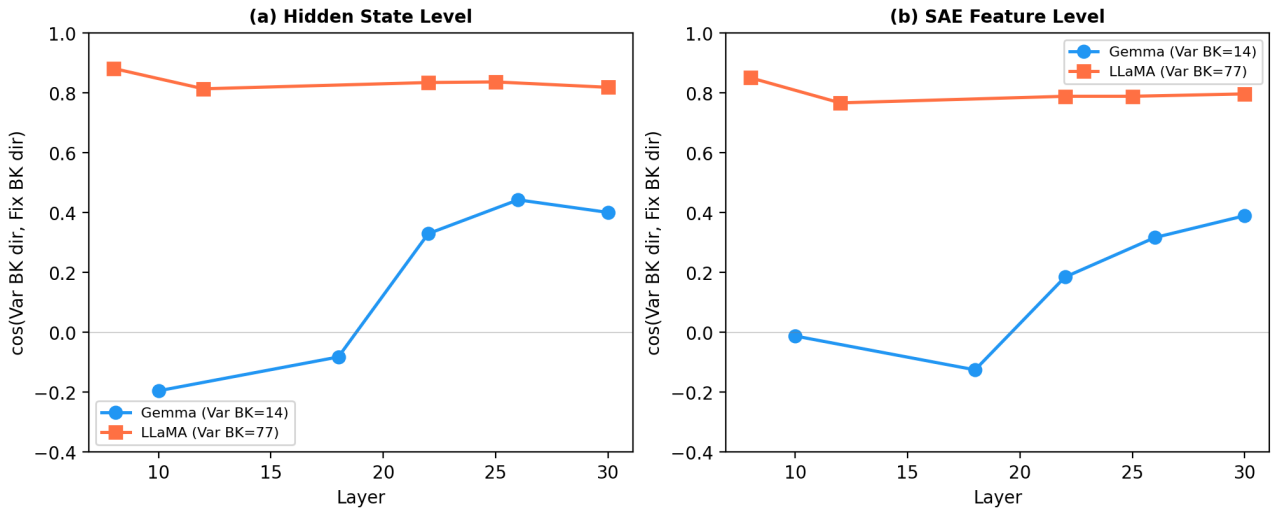

Figure 4: Var/Fix direction cosine at (a) hidden state and (b) SAE level. LLaMA cos$>$0.77 at all layers; Gemma converges at deep layers.

4.3 G-Prompt and Bet Constraint Effects

The G (Goal-setting) prompt shifts activations toward the BK direction in both models (cos = +0.85 in Gemma SM, +0.63 in LLaMA SM at L22). Bet constraints (c10$\rightarrow$c70) map linearly onto BK probability (Gemma r=0.979, LLaMA r=0.987; note n=4 data points limits the strength of this linear

claim). These observations suggest that the BK representation encodes continuous risk magnitude rather than a binary BK/Safe distinction.

4.4 RQ3 Summary

The F1 cross-bet-type transfer analysis provides direct evidence that Fixed and Variable betting conditions share the same underlying BK representation at deep layers. Direction cosines, common features, and G-prompt alignment all point in the same direction. BK encoding is robust to the Fixed/Variable manipulation in both models.

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5 Implications for NMT Paper §3.2

The paper's §3.2 identifies 112 causal features in LLaMA SM via activation patching, with safe features in L4–L19 and risky features in L24+. This analysis extends that single-model, single-domain finding in three ways:

Cross-model: Gemma achieves near-identical BK classification (0.976 vs 0.974) and exhibits the same balanced promoting/inhibiting neuron structure. The 112 causal features likely reflect a property present in both architectures, not a LLaMA-specific artifact.

Cross-domain: BK classifiers transfer across IC, SM, and MW with AUC up to 0.932. Factor decomposition shows 65–76% of features encode BK independently of paradigm. The causal features identified in SM likely generalize to other gambling domains.

Cross-condition: Cross-bet-type transfer (AUC 0.74–0.93) demonstrates that BK representations are robust to the Fixed/Variable manipulation—the same manipulation that produces the paper's central behavioral finding (Variable betting increases bankruptcy). The neural representation remains stable even as behavioral outcomes diverge.

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6 Limitations

1. **Correlational evidence only.** All analyses in this report are correlational. Causal validation via activation patching has been performed only in LLaMA SM (paper §3.2). Cross-model and cross-domain causal validation remains pending.

2. **Sample size imbalances.** Gemma IC has only 14 Variable BK cases, severely limiting power for Gemma bet-type analyses. Gemma MW has 54 BK cases. LLaMA MW has 2,426 BK (75.8%)—at this high base rate, even a trivial classifier performs well; balanced class weights partially mitigate this, but complementary metrics (balanced accuracy, F1) would strengthen the claim.

3. **Layer coverage.** LLaMA hidden states are extracted at 5 layers (L8, L12, L22, L25, L30), not all 32. Layer-specific phenomena between these checkpoints may be missed.

4. **Bet constraint linearity.** The r=0.979/0.987 linear mapping is computed from only 4 data points (c10, c30, c50, c70). With df=2, even random relationships frequently appear linear.

5. **Common BK features.** The 415 common features ($d \geq 0.3$ in both bet types, same sign) lack a permutation null comparison. The expected number of features meeting this criterion by chance has not been computed.

6. **Multiple comparisons.** FDR correction is applied within individual analyses but not across the full set. Effect sizes should be treated as primary evidence.

7. **Two models only.** Generalization to other architectures (e.g., Qwen, Mistral) has not been tested.

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7 Next Steps

1. **Gemma activation patching**: Test whether Gemma's 600 universal BK neurons causally influence BK outcomes, validating cross-model causal generalization.
2. **IC/MW activation patching**: Test causal features from SM in other paradigms.
3. **Permutation null for common features**: Compute expected number of features with $d \geq 0.3$ in both bet types under label permutation.
4. **Balanced accuracy for high-BK conditions**: Report balanced accuracy and F1 alongside AUC for LLaMA MW (75.8% BK).
5. **G-prompt mechanism**: Investigate the 16× behavioral gap (Gemma G: 20.75× vs LLaMA G: 1.26×) despite similar directional alignment.